Ahmad Mahmod

PhD in Computer Science - Networking

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Current Position

Enseignant - Chercheur (Assisstant Professor) *Lyon, France*

INSA Lyon & CITI Lab-Agora Team *Oct 2026 - Now*

Education

PhD in Computer Science *Strasbourg, France*

University of Strasbourg *Nov 2023 - Sep 2026*

- **Thesis**: Programmable and Self-adaptive Low-Power and Lossy Wireless Networks
- **Advisors**: Julien Montavont & Thomas Noël
- **Jury**: Adrien van den Bossche, Nathalie Mitton, Oana Iova, Cristel Pelsser
- **Grant:** MESR grant of the Doctoral School of Mathematics, Information Sciences and Engineering (MSII)

MSc in Telecommunication Engineering (GPA: 110 cum Laude/ 110) *Calabria, Italy*

University of Calabria *Oct 2021 - Oct 2023*

- ***First rank*** over the 2 years of the degree.
- **Thesis**: Improving the Quality of Federated Learning using Programmable Networks
- **Advisors**: Pasquale Pace & Antonio Iera
- **Grant:** International Master Scholarship of the University of Calabria

BS in Communications and Electronic Engineering (GPA: 93/100) *Latakia, Syria*

Latakia University (formerly: Tishreen University) *Sep 2014 - Sep 2019*

- ***First rank*** over the 5 years of the degree.
- **Thesis**: Millimeter Waves (mmWs) Channel Simulation using MATALAB
- **Advisor**: Ghadir Madi

Professional Experience

04/2019-08/2021	**Workshop Trainer in Telecommunications & Programming,** Skills and Career Center, Latakia University (formerly: Tishreen University), Syria
10/2019-08/2021	**Workshop Trainer in Telecommunications,** BIT Institute, Latakia, Syria

Publications

International Journals

Ahmad Mahmod, Julien Montavont, and Thomas Noel. 2026. FINE-STACK: Flexible architecture for low-power lossy IoT Networks with programmablE network STACK and semantic runtime updates. *Elsevier Ad Hoc Networks*.

Ahmad Mahmod, Julien Montavont, and Thomas Noel. 2026. PENSIL: ProgrammablE Network Stack for low-power lossy IoT networks using Lightweight-virtualization. *Elsevier Internet of Things*.

Ahmad Mahmod, Pasquale Pace, and Antonio Iera. 2024. The Role of SDN to Improve Client Selection in Federated Learning. *IEEE Communications Magazine*.

International Conferences

Ahmad Mahmod, Julien Montavont, and Thomas Noel. 2026. AI-Enabled Traffic-Aware Closed-Loop Framework for MAC-Layer Control in Multi-Hop IoT Networks. *The International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM).*

Ahmad Mahmod, Julien Montavont, and Thoms Noel. 2025. Programmable Solutions for Low-power Lossy Wireless Networks: A Study of SDN and Femto Containers. *International Conference on Advanced Information Networking and Applications (AINA)*.

Ahmad Mahmod, Pasquale Pace, and Antonio Iera. 2024. SDN-assisted client selection to enhance the quality of federated learning processes. *Proceedings of the IEEE Wireless Communications and Networking Conference (WCNC)*.

Ahmad Mahmod, Pasquale Pace, and Antonio Iera.. 2023. Improving the quality of federated learning processes via software defined networking. *NetAISys '23: Proceedings of the 1st International Workshop on Networked AI Systems*.

National Conferences

Ahmad Mahmod, Julien Montavont, and Thomas Noel. 2026. Divide and conquer: a semantic and modular reconfiguration of the network stack for constrained wireless networks. *CoRes 2026*. ***Best paper Award***.

Ahmad Mahmod, Julien Montavont, and Thomas Noel. 2024. Menu or a la carte? An architecture for programming the data plane of constrained wireless networks. *CoRes 2024*.

Teaching Experience

Wireless Networks — *Strasbourg, France*
University of Strasbourg - M1 (BAC+4) — *Spring 2025*

- Volume: 20h (Lectures (CM) + Exercises (TD) + Practical sessions (TP))
- Course Responsible: Ahmad Mahmod (me) & Fabrice Theoleyre

Internet of Things — *Strasbourg, France*
University of Strasbourg - M2 (BAC+5) — *Fall 2024 & Fall 2025*

- Volume: 36h (Practical sessions (TP))
- Course Responsible: Julien Montavont

Supervision

2026 **Charbel Gergès**, Master-1 Research Project (TER), *Reliable and Efficient Over-the-Air Software Update Mechanisms for Wireless Networks* (Bac+4). — *University of Strasbourg*

Awards & Grants

2026 **Best Paper Award**, CoRes 2026

2023 **Excellence Prize for First Graduate**, University of Calabria
2022 **Excellence Prize for First-ranked Student**, University of Calabria
2021 **International Master Scholarship**, University of Calabria

2019 **Excellence Certificate for First Graduate** , Latakia University
2018 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University
2017 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University
2016 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University
2015 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University

Certificates

2023 **Embedded Software and Hardware Architectures**, University of Colorado Boulder Online
2023 **Machine Learning Specialization**, Stanford Online
2022 **Deep Learning Specialization** , DeepLearning.AI
2020 **Google IT Automation with Python**, Google

Technical Skills

Programming	C/C++, Python, Java, MATLAB, Linux Bash
Machine Learning	scikit-learn, PyTorch, TensorFlow
SDN	RYU Controller, OpenDayLight Controller, POX Controller
Embedded Systems	RIOT, Contiki-NG, Zeypher
Tools	FIT IoT-LAB, Cooja, Mininet/Mininet-WiFi, NS3

Projects

Flexible Network Architecture for LLWNs
University of Strasbourg *2023–2026*

- **Used tools:** C/C++, Python, RIOT OS.
- **Code:** Repository 1 & Repository 2.

Closed-Loop ML-Based Reconfiguration of LLWNs
University of Strasbourg *2023–2026*

- **Used tools:** Machine learning, data analysis, C/C++, Python, RIOT OS.
- **Code:** Soon, pending publication acceptance.

RPL Non-Storing Mode Support in RIOT OS
University of Strasbourg *2025*

- **Used tools:** C/C++, Git.
- **Code:** Repository.

Improving Federated Learning Quality Using Programmable Networks
University of Calabria *2023*

- **Used tools:** Python, federated learning, GNS3, OpenDaylight Controller, OpenFlow.

Performance Evaluation of Shortest-Path Algorithms in SDN
University of Calabria *2023*

- **Used tools:** Ryu Controller, OpenFlow, NetworkX, Mininet, Dijkstra, Bellman-Ford, A*, Floyd-Warshall.
- **Code:** Repository.

Load Balancing for Face-Recognition Video Streaming
University of Calabria *2023*

- **Used tools:** Ryu Controller, OpenFlow, NetworkX, Mininet, Mininet-WiFi.
- **Code:** Repository.

RSNA Screening Mammography Breast Cancer Detection
University of Calabria *2023*

- **Used tools:** Machine learning, deep learning, computer vision, image processing, Python, TensorFlow, Kaggle.
- **Code:** Repository.

Contradiction and Entailment Detection in Multilingual Text

UNIVERSITY OF CALABRIA *2022*

- **Used tools:** Machine learning, natural language processing, deep learning, Python, TensorFlow, Hugging Face, Kaggle.
- **Code:** Repository.

Skin Disease Detector App

UNIVERSITY OF CALABRIA *2022*

- **Used tools:** Android Studio, Java, Python, TFLite, deep learning.
- **Code:** Repository.

Languages

ENGLISH	C1: Read, Written, Spoken
FRENCH	B2: Read, Written, Spoken
ITALIAN	B1: Spoken
ARABIC	Mother Tongue